

On Hierarchical Modeling of ICD Codes for EHR Foundation Models

Firstname Lastname

NAME@EMAIL.EDU

Department

University

City, State, Country

Firstname Lastname

NAME@EMAIL.EDU

Department

University

City, State, Country

Abstract

Electronic health record foundation models typically treat ICD diagnosis codes as flat tokens, overlooking the clinically meaningful hierarchical structure that captures disease families, subcategories, and fine-grained diagnostic detail. As a result, existing EHR representation learning methods do not explicitly exploit the hierarchical structure already present in the coding system. In this work, we study ICD hierarchy as a *general inductive bias* for clinical representation learning. We investigate two complementary mechanisms for incorporating hierarchy: first, by augmenting diagnosis sequences in a BERT-style transformer with tokens corresponding to different levels of the ICD hierarchy, and second, by injecting hierarchy into graph-based code representations through hierarchy-aware edges combined with diagnosis co-occurrence structure. Across these settings, we evaluate whether explicit hierarchy improves downstream prediction, which levels of the hierarchy are most useful, whether hierarchy encoding improves transfer across datasets, and how hierarchy reshapes embedding similarity structure. Our findings show that explicitly encoding ICD hierarchy improves over flat code representations in in-domain and cross-dataset settings, while revealing that the most useful level of hierarchy depends on both the task and the modeling approach. More broadly, we focus on hierarchy-aware EHR representation learning and show that the benefits of encoding hierarchy are generalizable across pretraining objectives and hierarchy levels .

1. Introduction

Electronic health records (EHRs) provide a longitudinal view of patient care through diagnoses, procedures, medications, and encounters. Recent transformer-based EHR models such as BEHRT and Med-BERT have shown that large-scale pretraining can learn useful patient representations for downstream clinical prediction tasks [Li et al. \(2020\)](#); [Rasmy et al. \(2021\)](#). However, much of this progress still treats structured clinical codes as flat symbols, even when those codes are drawn from ontologies that were explicitly designed to encode clinical hierarchy. This creates a mismatch between the structure of the data and the structure presented to the model. For instance, ICD codes are not arbitrary identifiers: they organize diseases into clinically meaningful families, subgroups, and increasingly

specific categories. [World Health Organization \(2004\)](#). A code, therefore, carries both a fine-grained diagnosis and a path through a broader clinical taxonomy. In practice, however, most EHR models represent ICD codes as independent tokens, ignoring this hierarchical structure. As a result, related diagnoses that should share information may instead be learned as isolated symbols. This is especially problematic in clinical data, where many diagnoses can be sparse, long-tailed, and institution-dependent.

We hypothesize that explicitly modeling hierarchy could benefit EHR representation learning in several ways. First, the hierarchy provides a natural inductive bias, so rare diagnoses can benefit from data associated with clinically similar codes. [Perotte et al. \(2014\)](#); [Xi et al.](#) Second, broader and intermediate groupings may capture care pathways or anatomical systems that can be more predictive than the most specific code alone [Song et al. \(2019\)](#). Third, hierarchy can improve robustness under *domain shift*, since higher-level disease families are often more stable across institutions [Ostrominski et al. \(2021\)](#). More broadly, encoding hierarchy can help learned embeddings better reflect clinical semantics rather than only empirical co-occurrence.

However, it remains unclear how hierarchy should be incorporated into modern EHR representation learning methods. One possibility is to inject hierarchy directly into the token representation used by a transformer encoder. Another is to encode hierarchy relationally through a graph over diagnosis codes, then learn hierarchy-aware code embeddings before downstream modeling. Prior work has shown that ontology structure can be useful for healthcare representation learning, including GRAM and G-BERT [Choi et al. \(2017\)](#); [Shang et al. \(2019\)](#), but these studies do not resolve how ICD hierarchy should be incorporated into transformer-based EHR foundation models or which level of hierarchy is most useful. A broad hierarchy may regularize the model but miss out on distinctions between diagnoses, whereas a very fine hierarchy may retain specificity but provide limited sharing.

Specifically, in this work, we make the following contributions:

- We study ICD hierarchy as an inductive bias for EHR foundation models and show that across different pretraining objectives, injecting ICD hierarchy helps in better representation learning.
- We propose two complementary hierarchy-aware modeling strategies: **HICD-BERT**, which injects hierarchy at the token-embedding level, and **HICD-Graph**, which injects hierarchy through a diagnosis graph and hierarchy-aware code embeddings.
- We perform an ablation study over broad, intermediate, and fine hierarchy levels, showing that different levels provide different benefits and, in general, hierarchy helps across tasks and pretraining tasks.
- We evaluate cross-dataset transfer to assess whether hierarchy-aware representations improve robustness under domain shift.
- We further analyse the embedding space to assess whether incorporating hierarchy results in more cohesive clustering of similar ICD codes.

2. Related Work

[This section is short and needs to be expanded – MT]

2.1. Sequential and Transformer-based EHR representation learning

Early work on longitudinal EHR modeling established the importance of learning from temporally ordered patient histories. Doctor AI [Choi et al. \(2016a\)](#) used recurrent neural networks to predict future clinical events from visit sequences, while RETAIN [Choi et al. \(2016c\)](#) introduced a reverse-time attention mechanism to improve interpretability in healthcare prediction. Dipole [Ma et al. \(2017\)](#) further demonstrated that attention-based sequence models can improve diagnosis prediction by capturing dependencies across visits. These methods showed that sequential structure in EHRs is highly informative, but they generally treated diagnosis codes as flat symbols.

More recently, Transformer-based models have become a central approach for structured EHR representation learning. BEHRT [Li et al. \(2020\)](#) was among the earliest and most influential frameworks to represent patient histories as sequences of medical codes and apply self-attention to capture dependencies across visits. Med-BERT [Rasmy et al. \(2021\)](#) further showed that masked pretraining on large EHR corpora can produce reusable representations for downstream prediction tasks. Hi-BEHRT [Li et al. \(2023\)](#) extends this paradigm by introducing hierarchical temporal windows to better capture multi-scale structure in patient trajectories. Despite these advances, most Transformer-based EHR models still treat diagnosis codes as atomic identifiers and do not explicitly encode the ontology structure of coding systems such as ICD.

2.2. Ontology-aware and hierarchy-aware medical representation learning

A parallel line of work has explored how medical ontologies can improve representation learning. GRAM [Choi et al. \(2017\)](#) is a foundational example, showing that incorporating ontology ancestors improves healthcare representations, particularly for infrequent concepts. This work introduced an important intuition that is central to our study: clinically related diagnoses should be able to share information through their position in a hierarchy. G-BERT [Shang et al. \(2019\)](#) extended this idea to graph-augmented pretraining, showing that structured medical knowledge can improve downstream clinical prediction.

More recent work has investigated hierarchy-aware modeling more explicitly. HiSGT [Zhou and Barbieri \(2025\)](#) encodes parent-child and sibling relations in the ICD taxonomy using a graph neural network, while MEDHMP [Wang et al. \(2023\)](#) emphasizes the value of capturing structure at multiple granularities during medical pretraining. These studies reinforce the importance of hierarchy as a source of clinical structure. However, they do not directly address the questions that motivate our work: how ICD hierarchy should be incorporated into Transformer-based EHR foundation models, whether graph-level or token-level integration is more effective, and which level of ICD hierarchy is most useful for downstream clinical prediction.

2.3. Graph-based Modeling of medical code structure

Graph-based methods provide another way to model relationships among medical concepts. Med2Vec [Choi et al. \(2016b\)](#) showed that co-occurrence-based embeddings can recover clinically meaningful code and visit representations even without explicit ontology supervision. More recent work has modeled dependencies among ICD codes as graphs to improve predictive performance. For example, LGG-NRGrasp [Xi et al.](#) demonstrates that modeling

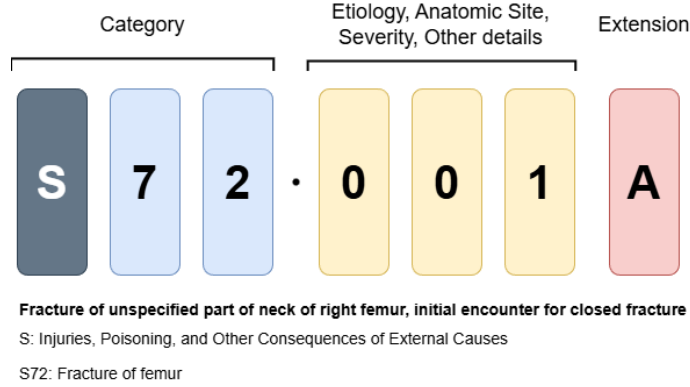


Figure 1: ICD code structure

structure in the diagnosis-label space can improve classification, particularly for sparse or rare codes.

These approaches are especially relevant to our graph-based formulation. However, much of the existing literature either focuses on graph learning alone or uses ontology structure for a specific downstream task. In contrast, we study ICD hierarchy as a broader inductive problem and ask not only whether graph structure helps, but whether it is a more effective mechanism for encoding ICD hierarchy than sequence-level augmentation.

3. Methods

We represent each patient as a temporally ordered sequence of visits, $\mathcal{V} = (v_1, \dots, v_T)$, where each visit v_t contains one or more ICD diagnosis codes. Our goal is to learn patient representations that preserve both temporal context and the hierarchical structure of diagnosis ontologies, and to evaluate whether such structure improves downstream prediction and generalization. We study two complementary mechanisms for injecting ICD hierarchy into EHR models. The first is a *token-level* approach, **HICD-BERT**, which extends BEHRT [Li et al. \(2020\)](#) by adding hierarchy-specific embedding tokens for each diagnosis token. The second is a *graph-level* approach, **HICD-Graph**, which constructs a hierarchy-augmented diagnosis co-occurrence graph, learns code embeddings with a graph convolutional network (GCN) [Kipf and Welling \(2017\)](#), and uses those embeddings to initialize a Transformer-based patient encoder. Full implementation details and hyperparameters are provided in [Appendix A](#).

3.1. ICD hierarchy levels

For each diagnosis code, we define three nested hierarchy levels, denoted $G0$, $G1$, and $G2$, ranging from coarse to fine. These levels are derived from the ICD-10 taxonomy and capture progressively more specific clinical groupings. Each ablation setting corresponds to enabling or disabling a subset of these levels; we evaluate all eight combinations $(G0, G1, G2) \in \{0, 1\}^3$, including the non-hierarchical baseline $(0, 0, 0)$.

Because the two models incorporate hierarchy through different mechanisms, the levels are operationalised differently in each. For **HICD-Graph**, hierarchy is encoded as explicit

graph nodes connected to leaf codes by weighted edges. $G0$ is the ICD-10 chapter (e.g. Chapter XIX, *Injury, poisoning and certain other consequences of external causes*), $G1$ is the official ICD-10-CM block (e.g. S70–S79, *Injuries to the hip and thigh*), and $G2$ is the 2-character prefix group (e.g. S7x, collecting all codes S70–S79). A leaf code such as S72.001A thus inherits $(G0, G1, G2) = (\text{Chapter XIX}, \text{S70–S79}, \text{S7x})$. For HICD-BERT, hierarchy is injected as auxiliary token embeddings alongside each leaf code token, using string-prefix truncation, a lightweight approach that requires no ontology access at training time. $G0$ is the 1-character prefix (e.g. S), $G1$ is the 2-character prefix (e.g. S7), and $G2$ is the 3-character ICD category (e.g. S72). The same leaf code S72.001A inherits $(G0, G1, G2) = (\text{S}, \text{S7}, \text{S72})$.

3.2. HICD-BERT : Hierarchy-aware BEHRT

HICD-BERT extends BEHRT Li et al. (2020), a Transformer encoder for longitudinal EHR sequences. In BEHRT, each diagnosis token is represented by the sum of the token, age, positional, and segment embeddings. We augment this representation with up to three hierarchy-specific embeddings:

$$\mathbf{h}_i^{(0)} = \mathbf{e}_{\text{code}}(c_i) + \mathbf{e}_{\text{age}}(a_i) + \mathbf{e}_{\text{pos}}(p_i) + \mathbf{e}_{\text{seg}}(s_i) + \mathbb{I}[G0]\mathbf{e}_{G0}(g_i^{(0)}) + \mathbb{I}[G1]\mathbf{e}_{G1}(g_i^{(1)}) + \mathbb{I}[G2]\mathbf{e}_{G2}(g_i^{(2)}), \quad (1)$$

where c_i is the diagnosis token, a_i is age, p_i is position, s_i is the segment identifier, and $g_i^{(0)}, g_i^{(1)}, g_i^{(2)}$ are the group identifiers at hierarchy levels $G0$, $G1$, and $G2$. This design preserves the original BEHRT tokenization while directly injecting the coarse-to-fine ICD structure into the embedding layer. The resulting sequence is processed by a stack of Transformer encoder blocks Vaswani et al. (2017). Each block contains multi-head self-attention, a position-wise feed-forward network, residual connections, and layer normalization. The hierarchy-aware embedding layer is the only change to the BEHRT backbone. Additional architectural and training details are provided in Appendix A.1.

We pretrain HICD-BERT using a masked language modeling (MLM) objective over the diagnosis sequence. Given a masked sequence $\tilde{\mathbf{x}}$, the model predicts the original diagnosis token at masked positions:

$$\mathcal{L}_{\text{MLM}} = - \sum_{i \in \mathcal{M}} \log p(c_i \mid \tilde{\mathbf{x}}), \quad (2)$$

where $\mathcal{M}$ denotes the set of masked positions.

For downstream prediction, we initialize the encoder from the pretrained MLM checkpoint and attach a task-specific multilayer perceptron to the final [CLS] representation. Binary tasks are optimized with binary cross-entropy with logits, while multiclass tasks are optimized with cross-entropy.

3.3. HICD-Graph : Graph-based Hierarchy Modeling

Our second model injects the ICD hierarchy at the code-graph level rather than the token-embedding level. We first construct a diagnosis co-occurrence graph over ICD codes, where horizontal edges reflect empirical co-occurrence within admissions and are weighted by pointwise mutual information (PMI):

$$\text{PMI}(a, b) = \log \frac{P(a, b)}{P(a)P(b)} + \epsilon, \quad (3)$$

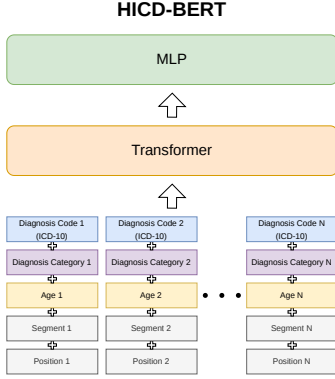


Figure 2: HICD-BERT

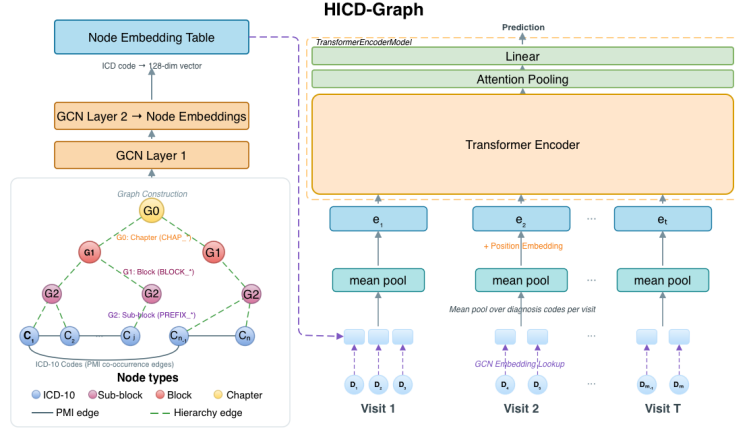


Figure 3: HICD-Graph

where $P(a)$ and $P(b)$ are marginal code frequencies and $P(a, b)$ is their joint co-occurrence probability. We retain only positive PMI edges above a percentile threshold to suppress weak associations. We then augment this graph with vertical hierarchy edges derived from the ICD taxonomy. [TODO: add more on ICD hierarchy encoding in the graph – MT]

Each diagnosis node can be connected to up to three ancestor nodes corresponding to the enabled hierarchy levels $G2$, $G1$, and $G0$. Horizontal PMI edges encode data-driven clinical relatedness, whereas vertical hierarchy edges encode ontology structure. This yields eight graph variants, one for each $(G0, G1, G2)$ combination. Given the resulting graph, we learn code embeddings using a two-layer GCN Kipf and Welling (2017) trained with a link-prediction objective:

$$\mathcal{L}_{\text{graph}} = - \sum_{(u,v) \in \mathcal{E}} \log \sigma(\mathbf{z}_u^\top \mathbf{z}_v) - \sum_{(u,v^-) \notin \mathcal{E}} \log (1 - \sigma(\mathbf{z}_u^\top \mathbf{z}_{v^-})), \quad (4)$$

where $\mathbf{z}_u$ and $\mathbf{z}_v$ denote node embeddings and $\mathcal{E}$ is the set of graph edges. The learned code embeddings are then used to initialize the embedding table of a patient-level Transformer encoder. For each visit, we aggregate the embeddings of its diagnosis codes and, when enabled, their hierarchy ancestors, to form a visit representation. The temporally ordered sequence of visits is then processed by a Transformer encoder, followed by attention-based pooling and a task-specific classification head. Additional implementation details are provided in Appendix A.2.

3.4. Experimental setup

All experiments are conducted on MIMIC-IV using a fixed 80/10/10 train/validation/test split shared across all hierarchy ablations. For HICD-BERT, we first pretrain the model with masked language modeling and then fine-tune it for each downstream task. For HICD-Graph, we first learn hierarchy-aware code embeddings from the diagnosis graph and then use these embeddings to initialize the downstream Transformer-based patient encoder. In HICD-BERT, we evaluate all subsets of the three hierarchy levels $G0$, $G1$, and $G2$, correspond to enabling

or disabling hierarchy embeddings; in **HICD-Graph**, they correspond to enabling or disabling the associated hierarchy edges and ancestor nodes.

We evaluate performance on **30-day readmission** and **next-visit-type prediction** using accuracy, F1, and AUROC. For transfer experiments, models trained on the source dataset (MIMIC) are evaluated on an external target dataset (eICU) without modifying the hierarchy construction pipeline. Exact optimization settings and hyperparameters are provided in Appendix A.

Task Definitions: [TODO – MT]

4. Results

We report the in-domain downstream performance of hierarchy-aware modeling, cross-dataset transfer under a linear-probe setting, and embedding-level analyses.

4.1. 30-day Readmission Task

Table 1 shows that adding hierarchical structure improves 30-day readmission prediction for both **HICD-BERT** and **HICD-Graph** relative to the no-hierarchy baseline. For **HICD-BERT**, we note that the strongest F1-macro is achieved by the two-level G0+G2 configuration (0.5953), while the best AUROC is obtained when all hierarchy levels are enabled (0.6923). This pattern suggests that hierarchical information is clearly beneficial overall, but that the gains are not strictly monotonic with increasing hierarchy information. For **HICD-Graph**, we note a more consistent pattern. The *all-levels* configuration, where all three–chapter, block and fine-grained group information is added, achieves both the best F1-macro (0.6031) and the best AUROC (0.7102), outperforming the no-hierarchy baseline as well as all partial-hierarchy variants. This suggests that the graph-based architecture is better able to exploit richer hierarchical structure when all levels are included simultaneously.

4.2. Binary Emergency-Visit Prediction

We report the results for binary emergency-visit prediction in Table 2. Here again, we note that hierarchy improves over the no-hierarchy baseline, but the pattern differs from the readmission task. For **HICD-BERT**, the best-performing configuration is the single-level G2 model –with only the finest ICD hierarchy token (e.g. S72) added– achieves the highest F1-macro (0.5200) and the highest AUROC (0.7941). For **HICD-Graph**, the strongest F1-macro is obtained by the two-level G0+G1 configuration (0.6443), while the best AUROC is achieved by the single-level G1 model (0.7582). Compared with **HICD-BERT**, **HICD-Graph** yields higher F1-macro across all configurations, whereas **HICD-BERT** attains the strongest AUROC overall. This contrast may reflect a difference in how the two architectures use hierarchical information. The graph-based model appears to produce more balanced thresholded predictions, while the token-based model appears to provide stronger ranking performance for this task. At the same time, unlike the readmission setting, enabling all hierarchy levels does not produce the best result; rather, single-level or two-level hierarchy variants perform better. This suggests that emergency-visit prediction depends more on a narrower level of diagnostic abstraction than on the full coarse-to-fine hierarchy. Regardless, all of the hierarchy-aware variants outperform the baseline deomnstraing the hierachcy of ICD code

Table 1: 30-day readmission results for HICD-BERT and HICD-Graph across hierarchy ablations.

Group	Hierarchy encoding	HICD-BERT		HICD-Graph	
		F1-macro	AUROC	F1-macro	AUROC
Baseline	No hierarchy	0.5786	0.6808	0.6026	0.6975
Single-level	G0	0.5825	0.6858	0.6010	0.7058
	G1	0.5740	0.6811	0.5892	0.7081
	G2	0.5896	0.6921	0.6014	0.7072
Two-Level	G0+G1	0.5696	0.6863	0.5990	0.7093
	G0+G2	0.5953	0.6918	0.5985	0.7095
	G1+G2	0.5834	0.6917	0.5889	0.7085
All-Levels	G0+G1+G2	0.5731	0.6923	0.6031	0.7102

is a powerful general inductive bias and should be included in ehr representation learning, showing that structured diagnosis information remains useful for this task as well.

4.3. Cross-Dataset Transfer Results

Table 3 presents cross-dataset transfer results on eICU under a linear-probe protocol. In this setting, the encoder pretrained on MIMIC-IV is kept frozen, and only a linear classifier is trained using eICU labels. We note that both HICD-BERT and HICD-Graph retain meaningful predictive signal under this evaluation setting, achieving competitive performance across hierarchy configurations. This indicates that the learned representations capture structure that transfers beyond the source corpus. For HICD-BERT, we note that the strongest transfer performance is achieved by the two-level G1+G2 configuration. HICD-Graph shows a similar overall pattern. Among the available configurations, the best F1-macro is shared by G1 and G0+G2, while G2 achieves the highest AUROC (0.584). Across both models, we note that the strongest transfer results are not obtained by enabling all hierarchy levels and encoding the full diagnosis relations. Instead, the best transfer performance is achieved by selective single-level or two-level variants. This suggests that, for cross-dataset transfer, a partial hierarchy, or single higher-level grouping such as G2, may offer a better balance between clinically meaningful abstraction and robustness to institutional differences.

4.4. Embedding analysis

For HICD-Graph, we analyze the learned *GCN node embeddings* specifically for leaf ICD diagnosis codes, to understand how hierarchical supervision changes the diagnosis-code representations. We study pairwise cosine similarity under three ICD relationship types: codes sharing the same 3-character ICD root, codes from the same ICD chapter but with different 3-character roots, and codes from different chapters. This setup allows us to distinguish whether hierarchy primarily sharpens very local diagnostic neighborhoods or induces broader structure in the embedding space.

Table 2: Binary emergency-visit prediction results for HICD-BERT and HICD-Graph across hierarchy ablations.

Group	Hierarchy encoding	HICD-BERT		HICD-Graph	
		F1-macro	AUROC	F1-macro	AUROC
Baseline	No hierarchy	0.4945	0.7682	0.6229	0.7508
Single-level	G0	0.4846	0.7848	0.6352	0.7540
	G1	0.5107	0.7698	0.6292	0.7582
	G2	0.5200	0.7941	0.6349	0.7569
Two-Level	G0+G1	0.5138	0.7838	0.6443	0.7574
	G0+G2	0.5159	0.7924	0.6174	0.7571
	G1+G2	0.5118	0.7903	0.6296	0.7580
All-Levels	GO+G1+G2	0.4988	0.7917	0.6234	0.7561

Table 3: CROSS DATASET TRANSFER: Linear Probe results on eICU readmission task for HICD-BERT and HICD-Graph across hierarchy ablations.

Group	Hierarchy encoding	HICD-BERT		HICD-Graph	
		F1-macro	AUROC	F1-macro	AUROC
Baseline	No hierarchy	0.470	0.569	0.380	0.564
Single-level	G0	0.473	0.566	0.396	0.576
	G1	0.475	0.568	0.487	0.579
	G2	0.459	0.570	0.461	0.584
Two-Level	G0+G1	0.482	0.569	0.467	0.582
	G0+G2	0.439	0.568	0.487	0.579
	G1+G2	0.487	0.571	0.453	0.574
All-Levels	GO+G1+G2	0.440	0.567	0.474	0.587

Figure 4(a) compares the mean cosine similarity across these three relationship types. Relative to the no-hierarchy baseline, the **all-levels** hierarchy config increases similarity in all three groups, with the largest increase observed for codes sharing the same 3-character ICD root. The resulting embedding space preserves a clinically meaningful similarity gradient: the closest ICD relatives are most similar, broader within-chapter relatives are less similar, and diagnoses from different chapters are least similar. This suggests that hierarchical supervision strengthens the local clinical organization of the embedding space rather than simply treating all diagnosis codes as independent identifiers.

Figure 4(b) examines how each different hierarchy configuration affects similarity among diagnoses sharing **the same 3-character ICD root**. All hierarchy-aware variants improve over the no-hierarchy baseline, showing that explicit ontology information consistently pulls closely related diagnoses closer together in embedding space. The strongest effect is obtained when the intermediate and fine hierarchy levels are combined (G1+G2), indicating that this configuration provides the most effective balance between specificity and information

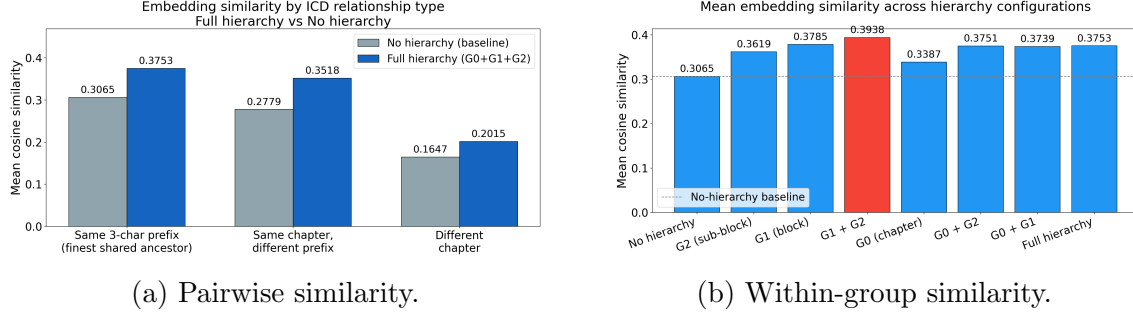


Figure 4: Embedding analysis for HICD-Graph. Full hierarchy increases similarity...

sharing. In contrast, the broadest hierarchy level alone (G0) yields smaller gains, suggesting that coarse grouping is less effective for tightening local diagnostic structure.

In Fig. 5 in the appendix, we have further added intra-chapter cosine similarity across diagnosis codes, and we observe that across individual chapters, cosine similarity increases with the addition of a hierarchy prior. However, the cosine similarity and the increase are heterogeneous across individual chapters.

Taken together, these analyses show that hierarchy in HICD-Graph improves the *local* clinical coherence of diagnosis embeddings, with the gains arising across all hierarchy level ablation.

5. Required: Discussion

[Add discussion on knowledge base graph, graog edges ane emb analyssi with similar ICD codes – MT]

This is probably the most important section of your paper! This is where you tell us how your work advances our understanding of machine learning and healthcare. Discuss both technical and clinical implications, as appropriate?.

Limitations Explain when your approach may not apply, or things you could not check. *Discussing limitations is essential. Both ACs and reviewers have been advised to be skeptical of any work that does not consider limitations.*

References

- Edward Choi, Mohammad Taha Bahadori, Andy Schuetz, Walter F Stewart, and Jimeng Sun. Doctor ai: Predicting clinical events via recurrent neural networks. In *Machine learning for healthcare conference*, pages 301–318. PMLR, 2016a.
- Edward Choi, Mohammad Taha Bahadori, Elizabeth Searles, Catherine Coffey, Michael Thompson, James Bost, Javier Tejedor-Sojo, and Jimeng Sun. Multi-layer representation learning for medical concepts. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 1495–1504, 2016b. doi: 10.1145/2939672.2939823.

- Edward Choi, Mohammad Taha Bahadori, Jimeng Sun, Joshua Kulas, Andy Schuetz, and Walter Stewart. Retain: An interpretable predictive model for healthcare using reverse time attention mechanism. *Advances in neural information processing systems*, 29, 2016c.
- Edward Choi, Mohammad Taha Bahadori, Andy Schuetz, Walter F. Stewart, and Jimeng Sun. Gram: Graph-based attention model for healthcare representation learning. In *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 787–795, 2017.
- Thomas N. Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. In *International Conference on Learning Representations*, 2017.
- Yikuan Li, Shishir Rao, José Roberto Ayala Solares, Abdelaali Hassaine, Rema Ramakrishnan, Dexter Canoy, Yajie Zhu, Kazem Rahimi, and Gholamreza Salimi-Khorshidi. Behrt: transformer for electronic health records. *Scientific reports*, 10(1):7155, 2020.
- Yikuan Li, Mohammad Mamouei, Gholamreza Salimi-Khorshidi, Shishir Rao, Abdelaali Hassaine, Dexter Canoy, Thomas Lukasiewicz, and Kazem Rahimi. Hi-BEHRT: Hierarchical Transformer-Based Model for Accurate Prediction of Clinical Events Using Multimodal Longitudinal Electronic Health Records. *IEEE Journal of Biomedical and Health Informatics*, 27(2):1106–1117, February 2023. ISSN 2168-2194, 2168-2208. doi: 10.1109/JBHI.2022.3224727. URL <https://ieeexplore.ieee.org/document/9964038/>.
- Fenglong Ma, Radha Chitta, Jing Zhou, Quanzeng You, Tong Sun, and Jing Gao. Dipole: Diagnosis prediction in healthcare via attention-based bidirectional recurrent neural networks. In *Proceedings of the 23rd ACM SIGKDD international conference on knowledge discovery and data mining*, pages 1903–1911, 2017.
- John W Ostrominski, Javier Amione-Guerra, Brian Hernandez, Joel E Michalek, and Anand Prasad. Coding variation and adherence to methodological standards in cardiac research using the national inpatient sample. *Frontiers in cardiovascular medicine*, 8:713695, 2021.
- Adler Perotte, Rimma Pivovarov, Karthik Natarajan, Nicole Weiskopf, Frank Wood, and Noémie Elhadad. Diagnosis code assignment: models and evaluation metrics. *Journal of the American Medical Informatics Association*, 21(2):231–237, 2014.
- Laila Rasmy, Yang Xiang, Ziqian Xie, Cui Tao, and Degui Zhi. Med-bert: pretrained contextualized embeddings on large-scale structured electronic health records for disease prediction. *npj Digital Medicine*, 4(1):86, 2021.
- Junyuan Shang, Tengfei Ma, Cao Xiao, and Jimeng Sun. Pre-training of graph augmented transformers for medication recommendation. *arXiv preprint arXiv:1906.00346*, 2019.
- Lihong Song, Chin Wang Cheong, Kejing Yin, William K Cheung, Benjamin CM Fung, and Jonathan Poon. Medical concept embedding with multiple ontological representations. In *IJCAI*, volume 19, pages 4613–4619, 2019.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, 2017.

Xiaochen Wang, Junyu Luo, Jiaqi Wang, Ziyi Yin, Suhan Cui, Yuan Zhong, Yaqing Wang, and Fenglong Ma. Hierarchical Pretraining on Multimodal Electronic Health Records, October 2023. URL <http://arxiv.org/abs/2310.07871>. arXiv:2310.07871 [cs].

World Health Organization. *ICD-10 : international statistical classification of diseases and related health problems : tenth revision*. World Health Organization, Geneva, 2nd edition, 2004. ISBN 9789241546492.

Suyang Xi, Jiesen Shi, MingJing Lin, Xinyi Zhou, Yuan Cheng, Hong Ding, and Chia Chao Kang. classification with a robust graph neural network for hierarchical coding.

Guanglin Zhou and Sebastiano Barbieri. Generating Clinically Realistic EHR Data via a Hierarchy- and Semantics-Guided Transformer, July 2025. URL <http://arxiv.org/abs/2502.20719>. arXiv:2502.20719 [cs].

Appendix A. Implementation Details

A.1. HICD-BERT Architecture and Training

HICD-BERT extends a BEHRT-style encoder by summing diagnosis-token, age, position, and segment embeddings, with optional additive embeddings for the three hierarchy levels G_0 , G_1 , and G_2 . The contextualized sequence is processed by a stack of Transformer blocks, and the pretraining objective is masked language modeling (MLM). For downstream prediction, the encoder is initialized from the pretrained checkpoint and a task-specific multilayer perceptron is applied to the final [CLS] representation.

A.1.1. ARCHITECTURE

HICD-BERT builds on a BEHRT-style Transformer encoder and augments the standard token, age, position, and segment embeddings with up to three hierarchy-aware embedding channels corresponding to G_0 , G_1 , and G_2 . These hierarchy embeddings are added to the input representation and are learned jointly with the rest of the model. For downstream prediction, the contextualized [CLS] representation is passed through a task-specific multilayer perceptron classifier.

Table 4: HICD-BERT architecture details.

Component	Setting
Backbone	BEHRT-style Transformer encoder
Embedding components	Token + age + position + segment + optional $G_0/G_1/G_2$ embeddings
Hidden size (d_{model})	512
Transformer layers	6
Attention heads	8
Feed-forward dimension	2048
Dropout	0.1
Maximum sequence length	256
MLM head	Linear $\rightarrow$ GELU $\rightarrow$ LayerNorm $\rightarrow$ tied decoder
Task head	3-layer MLP on [CLS] representation
Task output dimension	1 for binary tasks;

A.1.2. MLM PRETRAINING

We pretrain HICD-BERT with a masked language modeling objective over diagnosis sequences. The model is optimized to recover masked diagnosis tokens from their clinical context while jointly conditioning on age, position, segment, and the enabled hierarchy-level embeddings. Table 5 summarizes the pretraining configuration.

A.1.3. TASK FINE-TUNING

For downstream tasks, we initialize the encoder from the best MLM checkpoint and optimize a task-specific classification objective. Binary tasks use `BCEWithLogitsLoss` with a positive class weight $w^+ = N_{\text{neg}}/N_{\text{pos}}$ to address label imbalance, whereas multiclass next-visit-type prediction uses `CrossEntropyLoss` with inverse-frequency class weights. The best

Table 5: HICD-BERT MLM pretraining hyperparameters.

Hyperparameter	Value
Train batch size	256
Validation batch size	128
Number of workers	8
Epochs	100
Optimizer	AdamW
Learning rate	1×10^{-4}
Weight decay	0.01
Scheduler	CosineAnnealingLR
Scheduler $T_{\max}$	100
Minimum learning rate	1×10^{-6}
Loss	CrossEntropyLoss

checkpoint is selected by validation AUROC. For binary tasks, a decision threshold is swept over $[0.01, 0.99]$ (200 steps) on the validation set and stored in the checkpoint for test-set evaluation. Table 6 lists the full fine-tuning hyperparameters.

Table 6: HICD-BERT downstream fine-tuning hyperparameters.

Hyperparameter	Value
Train batch size	128
Validation batch size	256
Number of workers	8
Epochs	64
Optimizer	AdamW
Learning rate	1×10^{-5}
Weight decay	0.01
Scheduler	ReduceLROnPlateau (mode: max)
Scheduler patience	2
Scheduler factor	0.5
Early stopping patience	8
Early stopping min delta	10^{-4}
Early stopping monitor	Validation AUROC
Model selection criterion	Best validation AUROC
Binary-task loss	BCEWithLogitsLoss ($w^+ = N_{\text{neg}}/N_{\text{pos}}$)
Multi-class loss	CrossEntropyLoss (inverse-frequency weights)
Threshold selection	Sweep $[0.01, 0.99]$, 200 steps, max F1 on val set

A.2. HICD-Graph : Graph-Augmented Hierarchy Model

The graph-based model has two stages. First, we construct a diagnosis co-occurrence graph over ICD roots using PMI-weighted edges, then augment the graph with hierarchy edges corresponding to the enabled $G0$, $G1$, and $G2$ levels. Second, we train a two-layer GCN for node embedding learning and use the resulting embeddings to initialize a patient-level Transformer.

A.2.1. GRAPH CONSTRUCTION

We construct an undirected diagnosis graph whose nodes correspond to ICD root codes. Horizontal edges are weighted by PMI computed from within-admission co-occurrence, and vertical edges are added to encode enabled hierarchy levels. Table 7 summarizes the graph construction settings.

Table 7: Diagnosis graph construction hyperparameters.

Hyperparameter	Value
Diagnosis vocabulary	ICD-10-CM three-character root codes
ICD-9 handling	Mapped to ICD-10-CM root when available
Horizontal edges	PMI-weighted co-occurrence edges
PMI filtering	Positive PMI only
PMI cutoff	52nd percentile
Hierarchy levels	G_0 , G_1 , G_2 ablated independently
Hierarchy edge weight	1.0
Graph type	Undirected weighted graph

A.2.2. GCN EMBEDDING TRAINING

We learn node embeddings with a two-layer GCN trained using a link-prediction objective over observed and sampled non-edge pairs. The resulting embeddings are used to initialize the downstream patient encoder. The training configuration is given in Table 8.

Table 8: GCN embedding training hyperparameters.

Hyperparameter	Value
Input node feature dimension	64
Input feature initialization	Random normal
GCN layers	2
Hidden dimension	128
Output embedding dimension	128
Edge weights	Min-max normalized to $[0, 1]$
Training objective	Link prediction
Positive pairs	Existing graph edges
Negative pairs	Random non-edge pairs
Loss	BCEWithLogitsLoss
Optimizer	Adam
Learning rate	0.005
Epochs	400

A.2.3. TRANSFORMER FINE-TUNING ON GRAPH EMBEDDINGS

The learned graph embeddings initialize the diagnosis embedding table of a patient-level Transformer operating over temporally ordered visits. Visit representations are formed

by masked mean pooling over code embeddings, and patient representations are obtained through attention pooling across visits. Table 9 provides the fine-tuning details.

Table 9: Patient-level Transformer hyperparameters for the graph-based model.

Hyperparameter	Value
Embedding initialization	GCN node embeddings
Embedding dimension	128
Maximum visits per patient	10
Code aggregation within visit	Masked mean pooling
Visit positional encoding	Learnable
Transformer layers	2
Attention heads	4
Feed-forward dimension	256
Dropout	0.3
Pooling across visits	Attention pooling
Classifier	Linear layer on pooled patient representation
Batch size	64
Epochs	15
Optimizer	AdamW
Learning rate	2×10^{-5}
Weight decay	1×10^{-5}
Scheduler	ReduceLROnPlateau (mode: max)
Scheduler patience	2
Scheduler factor	0.5
Model selection criterion	Best validation AUROC
Loss	BCEWithLogitsLoss ($w^+ = N_{\text{neg}}/N_{\text{pos}}$)
Class balancing	WeightedRandomSampler + w^+
Threshold selection	Sweep [0.01, 0.99], 200 steps, max F1 on val set
Mixed precision	FP16 on CUDA
Gradient clipping	Max norm 1.0

A.3. Hierarchy Ablations

For both HICD-BERT and the graph-augmented model, we evaluate all eight hierarchy settings:

$$(0, 0, 0), (0, 0, 1), (0, 1, 0), (0, 1, 1), (1, 0, 0), (1, 0, 1), (1, 1, 0), (1, 1, 1).$$

Here, $(0, 0, 0)$ corresponds to the non-hierarchical baseline, while $(1, 1, 1)$ uses the full hierarchy. In HICD-BERT, these settings determine which hierarchy embeddings are added to the token representation. In the graph-based model, they determine which hierarchy edges are added to the diagnosis graph and which ancestor nodes are included during sequence tokenization.

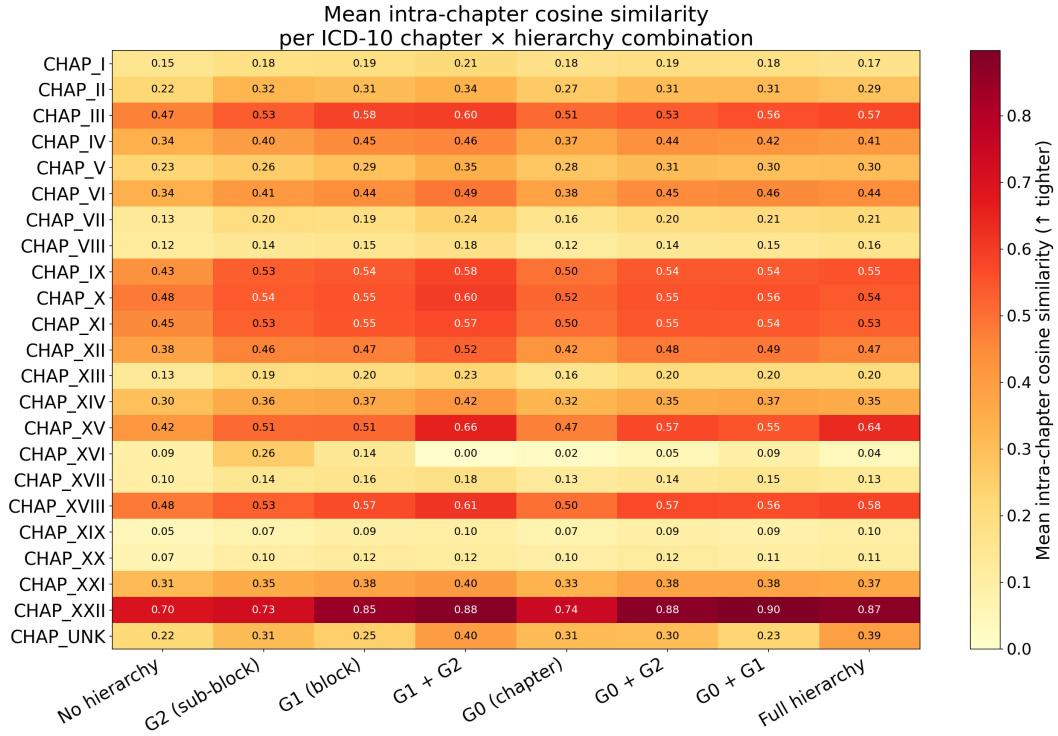


Figure 5: Mean pairwise cosine similarity among ICD-10 leaf codes within each chapter, computed across all eight hierarchy configurations (G0/G1/G2 ablations). Higher values indicate that codes belonging to the same chapter are embedded more tightly together.